| High Spin Complex | Low spin complex |
|---|---|
| Electronic configuration( for high spin complex) : $t_{2g}^3 e_g^2$<br>CFSE = {[3(-0.4)+2 (0.6)] $\Delta_o$ + 0 × P} - {0}<br>= 0 | Electronic configuration ( for low spin complex) : $t_{2g}^5 e_g^0$<br>CFSE = {[5(-0.4)+0 (0.6)] $\Delta_o$ + 2 × P} - {0}<br>= -2 $\Delta_o$ + 2P<br>= (-2 ×14000) + (2× 30000)<br>= 32000 $cm^{-1}$<br>High positive CFSE value indicates that low spin complex is not a favourable one. |
| Actual nature of $[Fe(H_2O)_6]^{3+}$ | High spin (Spin free) |
| Electronic configuration of central metal ion | $t_{2g}^3 e_g^2$ |
| Magnetic property | No. of unparied electrons = 5;<br>Hence paramagnetic |
| Magnetic moment<br>(Using spin only formula) | $\mu_s = \sqrt{n(n+2)} = \sqrt{5(5+2)} = 5.916$ BM |

**Complex: $[Fe(CN)_6]^{3-}$**

| High Spin Complex | Low spin complex |
|---|---|
| Electronic configuration in isotropic field : d5 ↑ ↑ ↑ ↑ ↑<br>No. of paired electrons (n'p)= 0 ; Therefore, $E_{iso}$ = 0 | |
| Ligand field:<br>Electronic configuration : $t_{2g}^3 e_g^2$<br>CFSE = {[3(-0.4)+2 (0.6)] $\Delta_o$ + 0 × P} - {0}<br>= 0 | Ligand field Electronic configuration : $t_{2g}^5 e_g^0$<br>CFSE = {[5(-0.4)+0 (0.6)] $\Delta_o$ + 2 × P} - {0}<br>= -2 $\Delta_o$ + 2P<br>= (-2 ×35000) + (2× 30000)<br>= -10000 $cm^{-1}$<br>Negative CFSE value indicates that low spin complex is favoured |



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